

Supplementary Material for “Parallel Multi-Stream Dual-Aggregation Unfolding Paradigm for Compressive Sensing Reconstruction”

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Overview

This supplementary material provides additional theoretical analyses and experiments to complement the main paper. Specifically, Section S1 presents a feature-domain consistency analysis of the proposed PMSI iteration with the original CS optimization problem. Section S2 interprets the DAIF operator as a learned proximal operator within the Plug-and-Play framework. Section S3 elaborates on the essential distinctions between PMD-Net and standard recurrent neural networks. Section S4 provides a contribution analysis of MDCT and SWST on different image types. Section S5 evaluates the generalization capability of PMD-Net under different sensing matrices. Section S6 presents ablation studies on DGF design choices and the necessity of physical measurement constraints. Section S7 analyzes the stability of the unfolding iteration, including empirical verification of learned step-size boundedness and per-stage data-consistency error. Finally, Section S8 provides a speed-quality trade-off visualization comparing PMD-Net against all baselines.

S1. Feature-Domain Consistency Analysis

A natural concern is whether iterating in the high-dimensional feature domain preserves consistency with the original CS optimization problem. We argue that this consistency is maintained through the design of the DGF unit and the encoder-decoder structure of PMD-Net.

Formally, let $E(\cdot)$ and $D(\cdot)$ denote the MSFE encoder and P-MSFD decoder, respectively. Define the feature-domain surrogate objective as:

$$\min_{\tilde{\mathbf{x}}} \frac{1}{2} \|D(\tilde{\mathbf{x}}) - \mathbf{x}^0\|_F^2 + \lambda R(\tilde{\mathbf{x}}), \quad (\text{S1})$$

where $R(\cdot)$ is the implicit regularizer induced by the DAIF operator and $\mathbf{x}^0 = F_S^\top(\mathbf{y}, \text{kernel} = \mathbf{A})$ is the initial estimate. Under the assumption that $D \circ E \approx \mathbf{I}$ —i.e., the encoder-decoder pair is approximately an isometry, as supported by the high reconstruction fidelity observed empirically—the objective in Eq. (S1) is first-order consistent with the original CS LASSO formulation in Eq. (2). This theoretical framing parallels the analysis adopted in FSOINet [1], which similarly operates a PGD-unrolled iteration in feature space.

Critically, the physical measurement constraint is not abandoned during feature-domain iteration. At every stage i , the DGF unit explicitly recomputes the measurement residual gradient $\mathbf{A}^\top(\mathbf{A}\mathbf{x}^{(i)} - \mathbf{y})$ in the image domain and injects it into the feature domain via Pixel Unshuffle, as formulated in Eq. (16). This operation is structurally analogous to the gradient descent step of the standard PGD algorithm (cf. Eq. (3)):

$$\underbrace{\mathbf{s}^{(i+1)-j}}_{\text{feature-domain gradient step}} \leftrightarrow \underbrace{\mathbf{r}^{(k)} = \mathbf{x}^{(k-1)} - \rho^{(k)} \Phi^\top(\Phi \mathbf{x}^{(k-1)} - \mathbf{y})}_{\text{PGD gradient step}}. \quad (\text{S2})$$

Because the measurement residual $\mathbf{A}^\top(\mathbf{A}\mathbf{x} - \mathbf{y})$ is recomputed from the current state estimate at every iteration, measurement consistency is actively enforced throughout the unrolled optimization rather than only at initialization. The complete iteration can thus be interpreted as solving Eq. (S1) via an approximate proximal gradient descent, inheriting the convergence properties of the PnP-PGD framework [2].

S2. DAIF as a Learned Proximal Operator

While DAIF does not admit a closed-form proximal mapping, it can be rigorously interpreted as a valid learned proximal operator within the Plug-and-Play (PnP) framework [3, 4]. Under the PnP paradigm, a denoiser or feature-transformation operator can substitute for the classical proximal step provided it satisfies

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non-expansiveness, i.e.,

$$\|\text{DAIF}(\mathbf{u}) - \text{DAIF}(\mathbf{v})\|_2 \leq \|\mathbf{u} - \mathbf{v}\|_2, \quad \forall \mathbf{u}, \mathbf{v}. \quad (\text{S3})$$

In DAIF, the channel-adaptive soft-thresholding operator $\text{Soft}(\cdot, \lambda_j)$ is inherently non-expansive by construction [5]. Furthermore, the use of LayerNorm and residual connections in both MDCT and SWST empirically constrains the Lipschitz constant of the overall operator, providing a practical approximation of Eq. (S3). Consequently, DAIF can be viewed as implicitly defining a learned regularizer $R(\tilde{\mathbf{x}})$, and the overall PMSI iteration converges to a fixed point under standard PnP convergence conditions [2, 6].

S3. Essential Distinction from Recurrent Neural Networks

Although PMD-Net adopts a recurrent-style, stage-by-stage architecture, it differs fundamentally from a standard RNN in four respects, as summarized in Table S1.

Table S1: Essential distinctions between PMD-Net (PMSI backbone) and a standard RNN.

Aspect	PMD-Net (PMSI)	Standard RNN
Physical constraint	Each stage recomputes the measurement residual $\mathbf{A}^T(\mathbf{A}\mathbf{x}^{(i)} - \mathbf{y})$, enforcing measurement fidelity at every iteration.	No explicit physical constraint; input is fixed and does not encode measurement fidelity.
Iteration semantics	Each stage has a clear optimization interpretation: DGF $\leftrightarrow$ gradient descent step; DAIF $\leftrightarrow$ proximal mapping step.	Hidden-state update has no corresponding optimization step; iterations lack physical meaning.
Parameter interpretation	Learnable thresholds λ_j correspond to sparsity regularization strengths; step sizes are physically interpretable.	Gate parameters (e.g., forget/input gates) carry no physical interpretation.
Convergence guarantee	Convergence to a fixed point is analyzable under the PnP-PGD framework [2, 4].	No convergence guarantee to a meaningful solution of a defined optimization problem.

In a standard RNN, the hidden state update takes the form $\mathbf{h}^{(k)} = f(\mathbf{h}^{(k-1)}, \mathbf{z})$, where $\mathbf{z}$ is a fixed input that does not change across iterations. In contrast, the DGF unit in PMD-Net computes a stage-dependent gradient injection based on the current reconstruction estimate $\mathbf{x}^{(i)}$, creating a physically-grounded feedback loop that drives the solution toward measurement consistency. This is precisely the mechanism that distinguishes a deep unfolding network from a black-box recurrent model: the iterative dynamics of PMD-Net are anchored to the CS measurement model at every stage, ensuring that the network cannot degenerate into an unconstrained sequence model even when operating entirely in feature space.

S4. Contribution Analysis of MDCT and SWST on Different Image Types

To analyze the distinct contributions of MDCT and SWST in DAIF, we evaluate Net-2 (w/o MDCT) and Net-3 (w/o SWST) from Section 5.1 on General100

(natural images with random structures, where channel-wise local modeling is critical) and Urban100 (urban scene images rich in periodic textures, where long-range spatial dependency modeling is essential).

Table S2: Comparison of PSNR (dB) / SSIM on General100 at $\tau \in \{0.01, 0.04, 0.10, 0.25\}$.

Method	$\tau = 0.01$		$\tau = 0.04$		$\tau = 0.10$		$\tau = 0.25$	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Net-2 (w/o MDCT)	23.65	0.6421	28.44	0.8136	32.96	0.9052	38.28	0.9614
Net-3 (w/o SWST)	23.78	0.6468	28.55	0.8177	33.07	0.9088	38.39	0.9636
PMD-Net	24.29	0.6543	29.10	0.8250	33.60	0.9160	38.95	0.9700

Table S3: Comparison of PSNR (dB) / SSIM on Urban100 at $\tau \in \{0.01, 0.04, 0.10, 0.25\}$.

Method	$\tau = 0.01$		$\tau = 0.04$		$\tau = 0.10$		$\tau = 0.25$	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Net-2 (w/o MDCT)	20.31	0.5542	24.38	0.7641	28.20	0.8743	33.17	0.9436
Net-3 (w/o SWST)	20.22	0.5511	24.29	0.7610	28.11	0.8712	33.08	0.9405
PMD-Net	20.58	0.5605	24.67	0.7702	28.51	0.8806	33.49	0.9488

As shown in Tab. S2 and S3, the two sub-modules exhibit clearly differentiated contributions. On General100, Net-2 (w/o MDCT) suffers a larger drop than Net-3 (w/o SWST) across all CS rates (e.g., 0.11 dB gap at $\tau = 0.10$), confirming that MDCT is more critical for natural images with irregular local structures. Conversely, on Urban100, Net-3 (w/o SWST) degrades more than Net-2 (e.g., 0.09 dB gap at $\tau = 0.10$), demonstrating that SWST is more critical for periodic textures. The qualitative comparisons in Fig. S2 and S3 further corroborate these findings. In both cases, the full PMD-Net achieves the best performance, confirming that MDCT and SWST provide complementary rather than redundant contributions.

S5. Generalization to Different Sensing Matrices

In PMD-Net and most existing CS methods, a learnable sensing matrix is adopted during training. To assess cross-matrix generalization, we replace the learnable matrix with three types of fixed alternatives—Gaussian $\mathcal{N}(0, 1/M)$, Orthogonal (QR decomposition), and DCT—and evaluate all compared methods on LIVE29 at $\tau = 0.10$.

Table S4: Comparison of PSNR (dB) / SSIM under learnable and fixed sensing matrices on LIVE29 at $\tau = 0.10$.

Sensing Matrix	DGU-Net ⁺		OCTUF		LTwIST		UFC-Net		MTC-CSNet		PMD-Net	
	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
Learnable	28.27	0.8379	28.23	0.8345	27.71	0.8233	27.88	0.8254	27.38	0.8207	28.75	0.8440
Gaussian	25.69	0.7659	25.68	0.7635	25.12	0.7503	25.31	0.7544	24.82	0.7487	26.15	0.7710
Orthogonal	26.04	0.7779	26.02	0.7755	25.45	0.7623	25.64	0.7654	25.13	0.7597	26.53	0.7860
DCT	25.86	0.7719	25.85	0.7695	25.26	0.7553	25.46	0.7594	24.97	0.7537	26.36	0.7800

As shown in Tab. S4, switching to a fixed matrix leads to performance drops across all methods, which is expected given that all models are jointly optimized

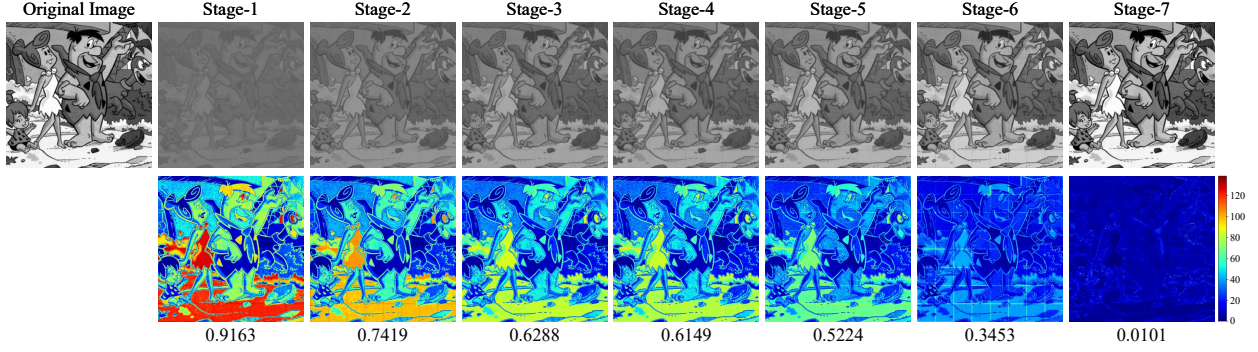


Figure S1: Per-stage reconstruction results and data-consistency error maps on a representative test image at $\tau = 0.10$. **Top row**: grayscale reconstructions. **Bottom row**: absolute error maps (jet colormap, unified scale) with normalized DC error below each stage, showing monotonic reduction from Stage 1 (DC = 0.9163) to Stage 7 (DC = 0.0101).

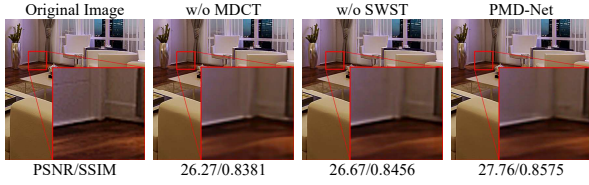


Figure S2: Visual comparison of reconstruction results on a representative natural image from General100 at $\tau = 0.10$.



Figure S3: Visual comparison of reconstruction results on a representative urban scene image from Urban100 at $\tau = 0.10$.

with their learnable matrices. Nevertheless, PMD-Net consistently achieves the best performance under every fixed matrix setting, outperforming the second-best method by 0.46 dB, 0.49 dB, and 0.50 dB under the Gaussian, Orthogonal, and DCT matrices, respectively. This consistent superiority demonstrates that PMD-Net’s reconstruction capability generalizes robustly across different sensing matrix types.

S6. Ablation on DGF Design Choices and Physical Measurement Constraints

To validate that the performance gains of PMD-Net stem from the principled injection of physical gradient information rather than from the mere addition of feature fusion operations, we construct four degraded variants: DGF-Cat (channel-wise concatenation with 1×1 projection), DGF-Add (learnable element-wise

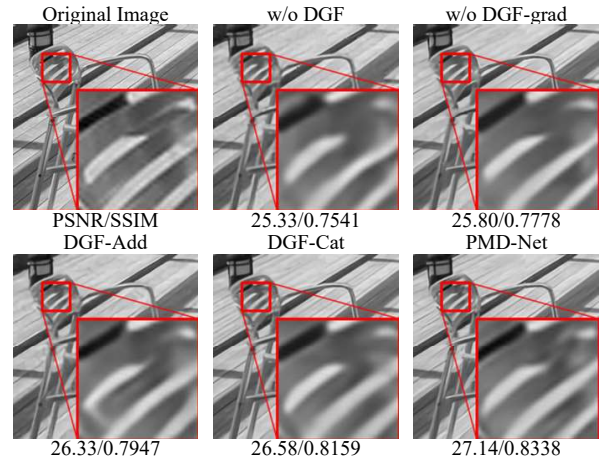


Figure S4: Visual comparison of reconstruction results under different DGF configurations at $\tau = 0.10$. Red boxes highlight zoomed-in regions for detailed comparison.

weighted summation), PMD-Net w/o DGF-grad (removing the $A^T y$ branch), and PMD-Net w/o DGF (removing the entire DGF unit).

Table S5: Comparison of PSNR (dB) / SSIM for different DGF configurations on Set11 and LIVE29 at $\tau \in \{0.01, 0.04, 0.10, 0.25\}$.

Method	Dataset	$\tau = 0.01$		$\tau = 0.04$		$\tau = 0.10$		$\tau = 0.25$	
		PSNR	SSIM	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
PMD-Net w/o DGF	Set11	21.14	0.5710	25.93	0.7871	29.64	0.8766	34.57	0.9336
	LIVE29	21.30	0.5143	24.15	0.6837	26.98	0.8087	31.62	0.9037
PMD-Net w/o DGF-grad	Set11	21.64	0.5813	26.45	0.8150	30.44	0.8991	35.89	0.9537
	LIVE29	21.87	0.5366	24.81	0.6970	28.07	0.8235	32.77	0.9150
DGF-Add	Set11	21.92	0.5902	26.81	0.8216	30.90	0.9045	36.21	0.9582
	LIVE29	22.25	0.5480	25.23	0.7102	28.36	0.8310	33.00	0.9245
DGF-Cat	Set11	22.13	0.5978	27.06	0.8263	31.22	0.9094	36.48	0.9611
	LIVE29	22.51	0.5553	25.50	0.7184	28.58	0.8382	33.16	0.9312
PMD-Net (Full)	Set11	22.35	0.6050	27.30	0.8300	31.50	0.9120	36.70	0.9640
	LIVE29	22.72	0.5615	25.75	0.7250	28.75	0.8440	33.30	0.9380

As shown in Tab. S5, a clear performance hierarchy emerges: Full > DGF-Cat > DGF-Add > PMD-Net w/o DGF-grad > PMD-Net w/o DGF. The gap between

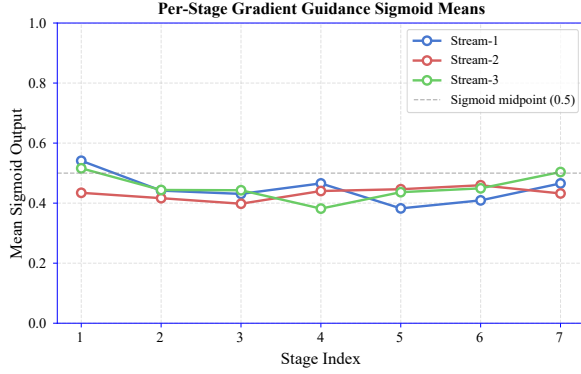


Figure S5: Per-stage mean sigmoid output of the three gradient guidance streams in DGF, averaged over Set11 at $\tau = 0.10$.

DGF-Cat/DGF-Add and the variants without physical inputs confirms that physical gradient information itself is indispensable. The further gap between DGF-Cat/DGF-Add and the full model (up to 0.43 dB on Set11 at $\tau = 0.01$) demonstrates that the deep feature extraction and sigmoid-guided weighting mechanism in DGF also play a positive role in maximally exploiting this information. The qualitative comparison in Fig. S4 corroborates these findings, showing progressive improvement in edge sharpness and structural detail preservation across configurations.

S7. Stability Analysis of Unfolding Iteration

Bounding of Learned Gradient Step Sizes. In PMD-Net, the gradient step size at each PMSI stage is realized as a sigmoid-activated guidance map generated by the DGF unit rather than an explicit scalar parameter. The sigmoid function $\sigma(\cdot)$ inherently constrains its output to $(0, 1)$, implicitly bounding the step size by construction. To verify that this constraint is active in the trained model, we measure the mean sigmoid output of all three gradient guidance streams across all seven PMSI stages, averaged over Set11 at $\tau = 0.10$. As shown in Fig. S5, all values remain stably within $(0.39, 0.55)$ throughout all stages without any drift toward the boundary values of 0 or 1, confirming that DGF learns well-regulated step sizes consistent with the convergence requirements of classical proximal gradient descent.

Data-Consistency Error per Stage. To further validate convergence behavior consistent with the unfolding claim, we measure the normalized DC error $\|\mathbf{Ax}^{(i)} - \mathbf{y}\|_2 / \|\mathbf{y}\|_2$ at each PMSI stage, where $\mathbf{x}^{(i)}$ is obtained by projecting the stage- i feature representation back to the image domain via the shared P-MSFD decoder. As

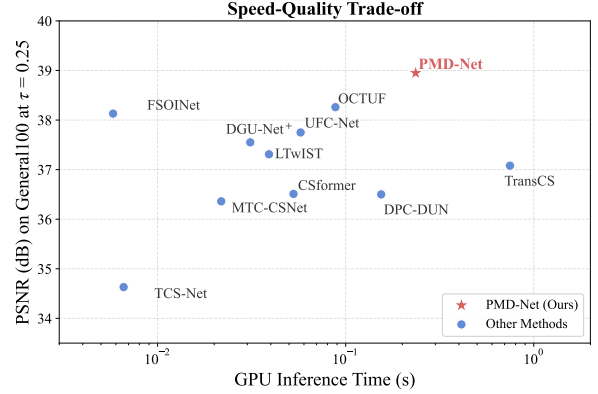


Figure S6: Speed-quality trade-off: GPU inference time vs. PSNR (dB) on General100 at $\tau = 0.25$. The horizontal axis uses a logarithmic scale. PMD-Net (red star) achieves the highest reconstruction quality while maintaining practical inference efficiency.

illustrated in Fig. S1, the DC error decreases monotonically from 0.9163 at Stage 1 to 0.0101 at Stage 7, a reduction of over 98%. The error maps corroborate this trend visually, with high-error regions in early stages progressively converging to near-uniform low-error maps by Stage 7, directly mirroring the iterative error reduction of classical PGD and confirming that PMD-Net implements a genuine convergent optimization trajectory.

S8. Speed-Quality Trade-off

To further illustrate the complexity trade-off discussed in Section 5.3, Fig. S6 presents a speed-quality trade-off plot of GPU inference time against PSNR on General100 at $\tau = 0.25$ for all compared methods. PMD-Net achieves the highest PSNR while maintaining a GPU inference time well within the range of computationally heavier baselines.

References

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